

Forcing monochromatic subdivisions

Gabriel Collado*

Dhruv Mubayi†

Abstract

We prove that for every integers $d \geq 1$ and $s \geq 2$ there exists an integer D , that depends only on d and s , such that for every graph P with maximum degree at most d , there is a graph H with maximum degree at most D in which every s -coloring of $V(H)$ yields a monochromatic subdivision of P .

1 Introduction

Ramsey theory studies the emergence of monochromatic substructures under arbitrary colorings. In its classical form, one considers edge-colorings of complete graphs and seeks monochromatic copies of a fixed graph. More precisely, the *classical Ramsey number* of a graph G , denoted $R(G; s)$, is defined as

$$R(G; s) = \min\{|V(K_n)| : K_n \xrightarrow{s} G\},$$

where $K_n \xrightarrow{s} G$ means that every coloring of $E(K_n)$ with s colors produces a monochromatic copy of G . The existence of such n for any G and s was established by Ramsey [12] and, independently, by Erdős and Szekeres [3]. When $H \xrightarrow{s} G$ we call G the *target graph* and H a *Ramsey host* for G . When $s = 2$ we omit the number of colors and write $R(G)$ for $R(G; 2)$ and $H \rightarrow G$ for $H \xrightarrow{2} G$.

One can ask the same question for any monotone graph parameter ρ . The ρ -*Ramsey number* of a graph G is defined as

$$R_\rho(G; s) = \min\{\rho(H) : H \xrightarrow{s} G\}.$$

This framework captures several well-studied variants. The classical Ramsey number corresponds to $\rho(H) = v(H)$, the number of vertices of H . Another natural choice is $\rho(H) = e(H)$, the number of edges of H , which gives the *size Ramsey number* $\hat{r}(G; s)$, introduced by Erdős, Faudree, Rousseau, and Schelp [2]. A third natural choice is $\rho(H) = \Delta(H)$, the maximum degree of H , which gives the *degree Ramsey number* $R_\Delta(G; s)$, studied systematically by Kinnersley, Milans, and West [9].

We focus on the case $\rho = \Delta$. The central question in this context is the following:

Question 1 ([6]). *Is $R_\Delta(G; s)$ bounded by a function of $\Delta(G)$ and s alone?*

Based on this question, we give the following definition:

*Department of Mathematics, Statistics and Computer Science, University of Illinois, Chicago, IL 60607. Email: gcoll8@uic.edu.

†Department of Mathematics, Statistics and Computer Science, University of Illinois, Chicago, IL 60607. Email: mubayi@uic.edu. Research partially supported by NSF Awards DMS-2153576 and DMS-2552740 and a Simons Fellowship.

Definition 2. A family $\mathcal{G}$ of graphs is said to be *degree-Ramsey bounded*, or R_Δ -bounded, if for every $s \in \mathbb{Z}^+$ and for every $G \in \mathcal{G}$ the degree-Ramsey number $R_\Delta(G; s)$ is bounded by a function depending only on $\Delta(G)$ and s .

Question 1 asks if the family of all graphs is R_Δ -bounded.

Several families are known to be R_Δ -bounded. Trees are R_Δ -bounded with the explicit bound $R_\Delta(T; s) \leq 2s(\Delta(T) - 1)$, as observed by Jiang. Cycles are also R_Δ -bounded: Jiang, Milans, and West [7] proved that $R_\Delta(C_n; 2)$ is at most 96 for even cycles and at most 3458 for odd cycles. More generally, Horn, Milans, and Rödl [6] proved that the family of closed blowups of trees is R_Δ -bounded. Recall that the *closed k -blowup* of a graph G , denoted $G[k]$, is obtained by replacing every vertex of G with a clique of size k and every edge with a complete bipartite graph $K_{k,k}$. This last result plus the fact that every graph of bounded treewidth and bounded maximum degree is a subgraph of a closed blowup of a bounded-degree tree imply that the family of graphs with bounded treewidth is R_Δ -bounded, as stated in Corollary 4.

Recall that a *tree decomposition* of a graph G is a tree $\mathcal{T}$ together with a collection of *bags* $\{B_t \subseteq V(G) : t \in V(\mathcal{T})\}$ satisfying the following three conditions: every vertex of G belongs to some bag; for every edge $uv \in E(G)$, some bag contains both u and v ; and for every vertex $v \in V(G)$, the set of nodes t with $v \in B_t$ induces a connected subtree of $\mathcal{T}$. The *width* of a tree decomposition is the size of its largest bag minus one, and the *treewidth* of G , denoted $\text{tw}(G)$, is the minimum width over all tree decompositions of G .

The following lemma states the fact that every graph of bounded treewidth and bounded maximum degree is a subgraph of a closed blowup of a bounded-degree tree in a precise manner.

Lemma 3 ([8]). *Every graph G with treewidth $\text{tw}(G) = w$ and maximum degree $\Delta(G) = d$ is a subgraph of the closed blowup $T[18wd]$ for some tree T of maximum degree at most $18wd^2$.*

Thus, the result of Horn, Milans, and Rödl [6] yields the following corollary.

Corollary 4. *The family of graphs with bounded treewidth is R_Δ -bounded. More precisely, for every integers $s \geq 2$, $d \geq 1$ and $w \geq 0$ there exists an integer D , depending only on s, d and w , such that for every graph P with maximum degree at most d and treewidth at most w , there is a graph H with maximum degree at most D such that every s -coloring of $E(H)$ yields a monochromatic copy of P .*

One can consider a natural variation of Ramsey numbers by coloring the *vertices* of the host graph rather than its edges. This is the direction we explore. We define the *vertex- ρ -Ramsey number* of G as

$$R_\rho^v(G; s) = \min\{\rho(H) : H \xrightarrow{s}_{\text{vtx}} G\},$$

where $H \xrightarrow{s}_{\text{vtx}} G$ means that every s -coloring of $V(H)$ contains a monochromatic copy of G as a subgraph in some color class. Analogously, we get the following definition:

Definition 5. We say a family $\mathcal{G}$ of graphs is *vertex- R_Δ -bounded* if for every $s \in \mathbb{Z}^+$ and for every $G \in \mathcal{G}$ the vertex-degree-Ramsey number $R_\Delta^v(G; s)$ is bounded by a function depending only on $\Delta(G)$ and s .

Vertex Ramsey theory has been studied since the work of Folkman [4] and Nešetřil, and Rödl [11], who established the existence of vertex Ramsey hosts of bounded clique number. The main open problem here is the natural analogue of Question 1.

Question 6. *Is $R_\Delta^v(G; s)$ bounded by a function of $\Delta(G)$ and s alone?*

Recall that a *subdivision* of a graph G is any graph obtained by replacing each edge of G with an internally vertex-disjoint path of length at least one. While we are not able to answer Question 6, our main result gives a positive answer once the required monochromatic target structure is enlarged from G itself to the collection of subdivisions of G . Our main result is the following:

Theorem 7. *For every integers $s \geq 2$ and $d \geq 1$ there exists an integer D , that depends only on d and s , such that for every graph P with maximum degree at most d , there is a graph H with maximum degree at most D such that every s -coloring of $V(H)$ yields a monochromatic subdivision of P .*

Our proof shows that blowups of bounded-degree expanders already suffice to enforce the desired vertex Ramsey property. The argument combines the expansion properties of a bounded-degree expander with a minor-universality theorem of Krivelevich and Nendov [10] to force a monochromatic minor of P inside some color class, and then uses a blowup construction to lift this minor to a subdivision of P .

We also establish vertex analogues of [6] and Corollary 4. Namely, we prove that the family of closed blowups of trees, and more generally the family of graphs with bounded treewidth, are both vertex- R_Δ -bounded. These results and proofs can be found in Section 5.

2 Notation and Preliminaries

Throughout, all graphs are finite and simple, and all logarithms are natural. For a graph F and a set $X \subseteq V(F)$, write

$$N_F(X) = \{y \in V(F) \setminus X : xy \in E(F) \text{ for some } x \in X\}$$

for the external neighborhood of X in F .

A *subdivision* of a graph P is obtained by replacing every edge of P by a path so that the replacing paths are internally vertex-disjoint. Equivalently, the original vertices of P become branch vertices, and two replacing paths may meet only at a common prescribed branch vertex. A graph P is a *minor* of a graph G if there are pairwise disjoint nonempty sets $W_v \subseteq V(G)$, indexed by $v \in V(P)$, such that every $G[W_v]$ is connected and, for each $uv \in E(P)$, there are vertices $x_u \in W_u$ and $x_v \in W_v$ such that $x_u x_v \in E(G)$. The sets W_v form a *minor model* of P in G .

For a positive integer s , write $[s] = \{1, \dots, s\}$, and let $B_s(G)$ denote the s -blowup of G : each vertex $x \in V(G)$ is replaced by an independent set

$$C_x = \{(x, 1), \dots, (x, s)\},$$

in $B_s(G)$ and, for each edge $xy \in E(G)$, all edges between C_x and C_y are added in $B_s(G)$.

If G is an r -regular graph with adjacency matrix $A(G)$ and the eigenvalues of $A(G)$ are $r = \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_N$. Define

$$\lambda(G) := \max_{2 \leq i \leq N} |\lambda_i|.$$

In other words, $\lambda(G)$ is the second largest eigenvalue in absolute value of $A(G)$.

Definition 8. Let $0 < \alpha < 1$ and $t \geq 1$. A graph F on q vertices is an (α, t) -expander if

$$|N_F(X)| \geq t|X| \quad \text{for every } X \subseteq V(F) \text{ with } |X| \leq \frac{\alpha q}{t}.$$

A graph is m -minor-universal if it contains every graph with at most m vertices and at most m edges as a minor.

3 Tools

We use the following minor-universality theorem of Krivelevich and Nenadov.

Theorem 9 (Krivelevich–Nenadov [10]). For every $0 < \alpha < 1$ there are constants $\xi > 0$ and $q_0, t_0 \in \mathbb{N}$ such that the following holds. If F is an (α, t) -expander on $q \geq q_0$ vertices and $t \geq t_0$, then F is m -minor-universal for

$$m = \xi \frac{q \log t}{\log q}.$$

We also use the existence of fixed-degree spectral expanders. Friedman’s second-eigenvalue theorem implies the following; see, for example, Bordenave [1].

Proposition 10. For every fixed integer $r \geq 3$ and every $\varepsilon > 0$, for all sufficiently large integers N with rN even there exists an N -vertex r -regular graph G such that

$$\lambda(G) \leq 2\sqrt{r-1} + \varepsilon.$$

For completeness, we record the form of the Expander Mixing Lemma that will be used.

Lemma 11. Let G be an N -vertex r -regular graph with $\lambda(G) \leq \lambda$. If $X, Z \subseteq V(G)$ are disjoint, then

$$\left| e_G(X, Z) - \frac{r}{N} |X| |Z| \right| \leq \lambda \sqrt{|X| |Z|},$$

where $e_G(X, Z)$ is the number of edges with one endpoint in X and the other in Z .

4 Proof of Theorem 7

We first extract a small-set expander from every large subset of a sufficiently strong spectral expander.

Lemma 12. Fix $s \in \mathbb{Z}^+$. Let $T \geq 100$ and put

$$\eta = \frac{1}{10s\sqrt{T}}.$$

Let G be an N -vertex r -regular graph satisfying $\lambda(G) \leq \eta r$. Then every set $S \subseteq V(G)$ with $|S| \geq N/s$ contains a set $U \subseteq S$ with $|U| > N/(3s)$ such that $G[U]$ is a $(1/(10s), T)$ -expander.

Proof. Set

$$a = \frac{N}{25sT} \quad \text{and} \quad b = \frac{N}{10sT}.$$

We first claim that

$$|N_{G[S]}(X)| \geq T|X| \tag{1}$$

for every $X \subseteq S$ satisfying $a \leq |X| \leq a + b$.

Suppose otherwise, and set

$$Y = N_{G[S]}(X), \quad Z = S \setminus (X \cup Y).$$

There are no edges between X and Z . Moreover,

$$|X| \leq a + b = \frac{7N}{50sT} \quad \text{and} \quad |Y| < T|X| \leq \frac{7N}{50s}.$$

Consequently,

$$|Z| \geq \frac{N}{s} - \frac{7N}{50sT} - \frac{7N}{50s} \geq \frac{N}{3s},$$

where the last inequality uses $T \geq 100$. By Lemma 11 and $e_G(X, Z) = 0$,

$$\frac{r}{N}|X||Z| \leq \lambda(G)\sqrt{|X||Z|}.$$

It follows that

$$|X||Z| \leq \frac{\lambda(G)^2}{r^2}N^2 \leq \eta^2N^2 = \frac{N^2}{100s^2T}.$$

Since $|Z| \geq N/(3s)$, this gives

$$|X| \leq \frac{3N}{100sT} < \frac{N}{25sT} = a,$$

a contradiction. This proves (1).

We now perform a pruning procedure. Start with $U_0 = S$. As long as there is a nonempty set $X_i \subseteq U_{i-1}$ such that

$$|X_i| \leq b \quad \text{and} \quad |N_{G[U_{i-1}]}(X_i)| < T|X_i|,$$

delete X_i and set $U_i = U_{i-1} \setminus X_i$.

We claim that the procedure deletes fewer than a vertices in total. Otherwise, let k be the first index for which

$$W = X_1 \cup \dots \cup X_k$$

has size at least a . By the minimality of k ,

$$a \leq |W| < a + b.$$

Furthermore,

$$N_{G[S]}(W) \subseteq \bigcup_{i=1}^k N_{G[U_{i-1}]}(X_i),$$

and therefore

$$|N_{G[S]}(W)| \leq \sum_{i=1}^k |N_{G[U_{i-1}]}(X_i)| < T \sum_{i=1}^k |X_i| = T|W|.$$

This contradicts (1).

Let U be the set remaining when the procedure terminates. Then

$$|U| > |S| - a \geq \frac{N}{s} - \frac{N}{25sT} > \frac{N}{3s}.$$

By the stopping rule,

$$|N_{G[U]}(X)| \geq T|X|$$

for every nonempty $X \subseteq U$ with $|X| \leq b$; the same inequality is trivial for $X = \emptyset$. Since

$$\frac{|U|}{10sT} \leq \frac{N}{10sT} = b,$$

the graph $G[U]$ is a $(1/(10s), T)$ -expander. $\square$

Recall that $B_s(P)$ denotes the s -blowup of P . The next lemma shows how to obtain a subdivision of a graph P from a minor of P by taking blowups.

Lemma 13. *Let s be a positive integer and let P be a graph with $\Delta(P) \leq s$. If P is a minor of G , then $B_s(G)$ contains a subdivision of P .*

Proof. Choose a minor model $\{W_v : v \in V(P)\}$ of P in G . For every edge $e = uv \in E(P)$, fix an edge of G between W_u and W_v , and write it as

$$x_{u,e}x_{v,e}, \quad x_{u,e} \in W_u, \quad x_{v,e} \in W_v.$$

For each $v \in V(P)$, choose a root $z_v \in W_v$ and an injection

$$\iota_v : \{e \in E(P) : e \text{ is incident with } v\} \longrightarrow [s].$$

For each edge e incident with v , choose a simple path in $G[W_v]$ from z_v to $x_{v,e}$, say

$$R_{v,e} = z_v = x_0, x_1, \dots, x_k = x_{v,e}.$$

Lift this path to $B_s(G)$ as

$$(z_v, 1), (x_1, \iota_v(e)), \dots, (x_k, \iota_v(e));$$

when $k = 0$, the lifted path consists only of $(z_v, 1)$. The lift is a path because consecutive clusters are joined by complete bipartite graphs.

For a fixed v , the lifted paths corresponding to distinct incident edges meet only at $(z_v, 1)$: away from the root they use distinct labels, and the underlying paths are simple. Lifted paths belonging to distinct vertices of P are disjoint because the branch sets W_v are disjoint.

For each edge $e = uv \in E(P)$, join the terminal vertex of the lifted path for (u, e) to the terminal vertex of the lifted path for (v, e) . This edge exists because $x_{u,e}x_{v,e} \in E(G)$ and the corresponding clusters form a complete bipartite graph in G . Concatenating the two lifted paths with this joining edge gives the path that replaces e .

Let J be the subgraph consisting of all root vertices $(z_v, 1)$, all lifted paths, and all joining edges constructed above. In particular, if v is isolated in P , then $(z_v, 1)$ is retained as an isolated branch vertex of J . The replacing paths are internally vertex-disjoint and meet precisely at their prescribed branch vertices. Hence J is a subdivision of P in $B_s(G)$. $\square$

We proceed now with the proof of our main theorem.

Proof of Theorem 7. Fix integers $d \geq 1$ and $s \geq 2$. Apply Theorem 9 with $\alpha = 1/(10s)$, and let $\xi > 0$ and $q_0, t_0 \in \mathbb{N}$ be the resulting constants. Take

$$T = \max\{100, t_0\}$$

and put

$$\eta = \frac{1}{10s\sqrt{T}}.$$

Choose an integer $r \geq 3$ sufficiently large such that

$$2\sqrt{r-1} + 1 \leq \eta r. \quad (2)$$

Define

$$D = sdr.$$

Notice that this number depends only on d and s .

Let P be any graph with $\Delta(P) \leq d$, and set

$$m_P = \max\{|V(P)|, |E(P)|\}.$$

Thus P is among the graphs covered by m_P -minor-universality.

Choose an integer N sufficiently large, subject to the parity condition rN even, so that Proposition 10 applies and

$$N \geq 6s, \quad \frac{N}{3s} \geq q_0, \quad \text{and} \quad \xi \frac{N \log T}{3s \log N} \geq m_P. \quad (3)$$

Such admissible integers N are unbounded, and $N/\log N \rightarrow \infty$ along them. Proposition 10, with $\varepsilon = 1$, therefore gives an N -vertex r -regular graph G such that

$$\lambda(G) \leq 2\sqrt{r-1} + 1 \leq \eta r.$$

Let

$$H = B_{sd}(G).$$

Every vertex of H has sd neighbors in each of the r clusters corresponding to its neighbors in G . Hence

$$\Delta(H) = sdr = D.$$

Consider an arbitrary s -coloring of $V(H)$. For each $x \in V(G)$, assign to x a majority color of its cluster C_x , breaking ties arbitrarily. At least N/s vertices of G receive the same assigned color. Without loss of generality, that color is red; let S be the set of vertices in $V(G)$ assigned red. Thus $|S| \geq N/s$, and every cluster C_x with $x \in S$ contains at least d red vertices.

By Lemma 12, there is a set $U \subseteq S$ with $q := |U| > N/(3s)$ such that

$$Q := G[U]$$

is a $(1/(10s), T)$ -expander. Since $N \geq 6s$, we have $q > N/(3s) \geq 2$, so all logarithms below are positive. Moreover, $q \leq N$ and $q > N/(3s)$. Hence,

$$\xi \frac{q \log T}{\log q} \geq \xi \frac{N \log T}{3s \log N} \geq m_P.$$

Also $q \geq q_0$ and $T \geq t_0$. Theorem 9 therefore implies that Q contains P as a minor.

For each $x \in U$, choose exactly d red vertices from the cluster C_x . The subgraph induced by all the chosen vertices is a red copy of $B_d(Q)$. Since $\Delta(P) \leq d$ and P is a minor of Q , Lemma 13 gives a subdivision of P inside this red copy of $B_d(Q)$. Thus all vertices of the subdivision are red and this finishes the proof of the theorem. $\square$

5 Vertex Ramsey Theory for Trees and Bounded-Treewidth Graphs

In this section we prove that the families of trees, closed blowups of trees and graphs with bounded treewidth are vertex- R_Δ -bounded.

We begin by recording the tree-embedding theorem of Friedman and Pippenger, which is the main tool that we will use.

Theorem 14 (Friedman–Pippenger [5]). *Let $d \geq 1$ and $n \geq 1$. If H is a non-empty graph satisfying*

$$|N_H(X)| \geq (d+1)|X|$$

for every $X \subseteq V(H)$ with $|X| \leq 2n-2$, then H contains every tree with at most n vertices and maximum degree at most d .

Theorem 15. *The family of trees is vertex- R_Δ -bounded. More precisely, for every integers $d \geq 1$ and $s \geq 2$ there is an integer D , depending only on d and s , such that for every tree T with $\Delta(T) \leq d$ there is a graph H with $\Delta(H) \leq D$ in which every s -coloring of $V(H)$ contains a monochromatic copy of T .*

Proof. Fix integers $d \geq 1$ and $s \geq 2$. Set $T_0 = \max(d+1, 100)$ and put

$$\eta = \frac{1}{10s\sqrt{T_0}}.$$

Choose an integer $r \geq 3$ large enough such that $2\sqrt{r-1} + 1 \leq \eta r$, and set $D = r$. Notice that D depends only on d and s .

Let T be any tree with $\Delta(T) \leq d$ and $n = |V(T)| > 1$ (if $n = 1$ the result is trivial). Take $N \geq 30s^2(2n-2)T_0$ sufficiently large with rN even. By Proposition 10 with $\varepsilon = 1$, there exists an N -vertex r -regular graph G with

$$\lambda(G) \leq 2\sqrt{r-1} + 1 \leq \eta r.$$

Consider an arbitrary s -coloring of $V(G)$. At least N/s vertices receive the same color; without loss of generality that color is red. Let S be the set of red vertices, so $|S| \geq N/s$. By Lemma 12, there is a set $U \subseteq S$ with $q := |U| > N/(3s)$ such that $G[U]$ is a $(1/(10s), T_0)$ -expander.

We verify the hypothesis of Theorem 14. Since

$$q > \frac{N}{3s} \geq 10s(2n-2)T_0,$$

we have $q/(10sT_0) > 2n-2$. Hence for every $X \subseteq V(G[U])$ with $|X| \leq 2n-2$, the $(1/(10s), T_0)$ -expansion condition applies and gives

$$|N_{G[U]}(X)| \geq T_0|X| \geq (d+1)|X|.$$

By Theorem 14, $G[U]$ contains T as a subgraph. Since every vertex of $G[U]$ belongs to S , this is a monochromatic copy of T . $\square$

Using the majority-coloring strategy from the proof of Theorem 7, we extend this result to closed blowups of trees.

Corollary 16. *The family of closed blowups of trees is vertex- R_Δ -bounded. More precisely, for every integers $d \geq 1$, $s \geq 2$ and $k \geq 1$ there is an integer D , depending only on d , s and k , such that for every tree T with $\Delta(T) \leq d$ there is a graph H with $\Delta(H) \leq D$ in which every s -coloring of $V(H)$ contains a monochromatic copy of the closed k -blowup $T[k]$.*

Proof. Fix integers $d, k \geq 1$ and $s \geq 2$. Let r , T_0 , and η be the constants from the proof of Theorem 15, and set

$$D = sk(r+1) - 1.$$

This depends only on d, k and s .

Let T be any tree with $\Delta(T) \leq d$ and $n = |V(T)|$. Choose $N \geq 30s^2(2n-2)T_0$ with rN even, and let G be an N -vertex r -regular graph with $\lambda(G) \leq \eta r$, as given by Proposition 10. Set $H = G[sk]$, the closed sk -blowup of G . Each vertex of H lies in a clique C_x of size sk and is adjacent to all sk vertices in each of the r neighboring clusters, so

$$\Delta(H) = (sk-1) + skr = sk(r+1) - 1 = D.$$

Consider an arbitrary s -coloring of $V(H)$. For each $x \in V(G)$, assign to x the majority color in its cluster C_x , breaking ties arbitrarily. At least N/s vertices of G receive the same majority color; without loss of generality that color is red. Let $S \subseteq V(G)$ be the set of majority-red vertices. Every cluster C_x with $x \in S$ contains at least k red vertices.

By Lemma 12 and the argument of Theorem 15, there is a set $U \subseteq S$ with $|U| > N/(3s)$ such that $G[U]$ is a $(1/(10s), T_0)$ -expander, and Theorem 14 yields an embedding $\phi: V(T) \rightarrow U$ of T in $G[U]$.

For each $v \in V(T)$, choose exactly k red vertices from $C_{\phi(v)}$; call this set R_v . The sets $\{R_v : v \in V(T)\}$ are pairwise disjoint since ϕ is injective. Since $C_{\phi(v)}$ is a clique of size sk in H , the set R_v induces a copy of K_k . For each edge $vw \in E(T)$, the edge $\phi(v)\phi(w)$ lies in $E(G)$, so all edges between $C_{\phi(v)}$ and $C_{\phi(w)}$ are present in H ; in particular, R_v and R_w span a copy of $K_{k,k}$. Therefore the subgraph of H induced by $\bigcup_{v \in V(T)} R_v$, together with all edges between R_v and R_w for $vw \in E(T)$, is a monochromatic red copy of $T[k]$. $\square$

Recall that every graph of bounded treewidth and bounded maximum degree is a subgraph of the closed blowup of a bounded-degree tree (Lemma 3). Together with Corollary 16, this gives the following.

Corollary 17. *The family of graphs with bounded treewidth is vertex- R_Δ -bounded. More precisely, for every integers $s \geq 2$, $d \geq 1$ and $w \geq 0$ there exists an integer D , depending only on s, d and w , such that for every graph P with maximum degree at most d and treewidth at most w , there is a graph H with maximum degree at most D such that every s -coloring of $V(H)$ yields a monochromatic copy of P .*

6 Concluding remarks

One can consider the vertex Ramsey problem with $\rho(G) = \chi(G)$, where $\chi(G)$ is the chromatic number of G . Perhaps surprisingly, this turns out to be rather straightforward as we observe below.

Proposition 18. *Let $K_{c;n}$ be the complete c -partite graph with parts of size n . Then*

$$R_\chi^v(K_{c;n}) = 2c - 1.$$

Proof. If $c = 1$ the result is trivial. Let then $c \geq 2$. For the lower bound, suppose that H is a graph with $\chi(H) \leq 2c - 2$. Consider an optimal coloring of H with at most $2c - 2$ colors and color half of the color classes red and the other half blue. Since each color class induces a graph with chromatic number at most $c - 1$, there is no monochromatic copy of $K_{c;n}$. This shows $R_\chi^v(K_{c;n}) \geq 2c - 1$. To see $R_\chi^v(K_{c;n}) \leq 2c - 1$, observe that $K_{2c-1;2n} \xrightarrow{2}_{vtx} K_{c;n}$ as the pigeonhole principle implies that in any 2-coloring of $V(K_{2c-1;2n})$, one of the colors has at least n vertices in at least c classes. $\square$

The proof of the general version with s colors is analogous and gives us the following result.

Proposition 19. *Let $K_{c;n}$ be the complete c -partite graph with parts of size n . Then*

$$R_\chi^v(K_{c;n}; s) = s(c - 1) + 1.$$

If G is any n -vertex graph with $\chi(G) = c$, then $G \subseteq K_{c;n}$ and this implies $R_\chi^v(G; s) \leq s(c - 1) + 1$, by Proposition 19. On the other hand, if $\chi(H) \leq s(c - 1)$ then its vertices can be partitioned into sets of size s , each inducing a graph of chromatic number at most $c - 1$, so none can contain G . Thus, $R_\chi^v(G; s) = s(c - 1) + 1$. Moreover, this generalizes naturally to k -uniform hypergraphs. Indeed, replacing the underlying graph structure with a k -uniform hypergraph $\mathcal{H}$, an analogous argument via the pigeonhole principle yields $R_\chi^v(K_{c;n}^{(k)}; s) = s(c - 1) + 1$, where $K_{c;n}^{(k)}$ denotes the complete c -partite k -uniform hypergraph with parts of size n .

The n by n square grid is the graph $P_n \square P_n$. One barrier to addressing Question 6 is that we could not determine the answer for the grid graph. We end with this question.

Question 20. *Is the family of square grids vertex- R_Δ -bounded?*

Declaration of AI Assistance

ChatGPT pro was used to assist in producing the technical details in the proof of Lemma 12. All other proofs in the paper, including the idea to use blowups of expanders and the Krivelevich-Nenadov result (Theorem 9) to obtain minors and then to apply Lemma 13 to obtain subdivisions were due to the authors. The authors take full responsibility for the content of the paper, including all mathematical statements, proofs, and references.

References

- [1] Charles Bordenave. A new proof of Friedman’s second eigenvalue theorem and its extension to random lifts. *Ann. Sci. Éc. Norm. Supér. (4)*, 53(6):1393–1439, 2020.
- [2] P. Erdős, R. J. Faudree, C. C. Rousseau, and R. H. Schelp. The size Ramsey number. *Period. Math. Hungar.*, 9(1-2):145–161, 1978.
- [3] P. Erdős and G. Szekeres. A combinatorial problem in geometry. *Compositio Math.*, 2:463–470, 1935.
- [4] Jon Folkman. Graphs with monochromatic complete subgraphs in every edge coloring. *SIAM J. Appl. Math.*, 18:19–24, 1970.
- [5] J. Friedman and N. Pippenger. Expanding graphs contain all small trees. *Combinatorica*, 7(1):71–76, 1987.
- [6] Paul Horn, Kevin G. Milans, and Vojtěch Rödl. Degree Ramsey numbers of closed blowups of trees. *Electron. J. Combin.*, 21(2):Paper 2.5, 6, 2014.
- [7] Tao Jiang, Kevin G. Milans, and Douglas B. West. Degree Ramsey numbers for cycles and blowups of trees. *European J. Combin.*, 34(2):414–423, 2013.
- [8] Nina Kamcev, Anita Liebenau, David R. Wood, and Liana Yepremyan. The size Ramsey number of graphs with bounded treewidth. *SIAM J. Discrete Math.*, 35(1):281–293, 2021.
- [9] William B. Kinnersley, Kevin G. Milans, and Douglas B. West. Degree Ramsey numbers of graphs. *Combin. Probab. Comput.*, 21(1-2):229–253, 2012.
- [10] Michael Krivelevich and Rajko Nenadov. Minors in small-set expanders. *Proc. Amer. Math. Soc.*, 154(4):1407–1420, 2026.
- [11] Jaroslav Nešetřil and Vojtěch Rödl. The Ramsey property for graphs with forbidden complete subgraphs. *J. Combinatorial Theory Ser. B*, 20(3):243–249, 1976.
- [12] F. P. Ramsey. On a Problem of Formal Logic. *Proc. London Math. Soc. (2)*, 30(4):264–286, 1929.